# Identifying Heterogeneity by Time Between Shocks

Exploiting exogenous variation in cyclone arrival timing

# Outline


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Strategy 1 — Hazard Residualization
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- Method
- Empirical results
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# 1. The Problem

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## 1.1 Baseline and current specification

# Baseline specification (context, no gap)

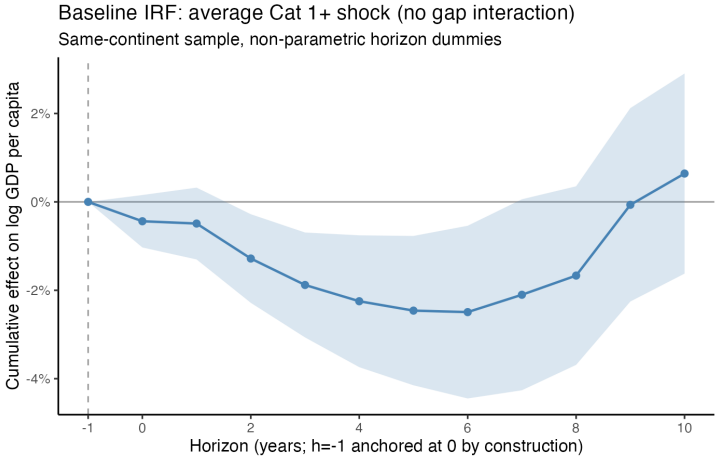
Before introducing the gap interaction, the workhorse LP estimates the *average* effect of a Cat 1+ shock (64 knots, $n = 374$ events):

$$y_{i,t+h} - y_{i,t-1} \;=\; \beta(h)\,\mathrm{shock}_{it} \;+\; \gamma_h' X_{it} + \alpha_{ih}(t) + \delta_{th} + \varepsilon_{ith}$$

where $\beta(h)$ is estimated non-parametrically via horizon dummies (one coefficient per $h \in \{0, 1, \ldots, H\}$), $X_{it}$ = two lags of GDP growth and shock, $\alpha_{ih}(t)$ = country$\times h$ FE with linear + quadratic trends, $\delta_{th}$ = year$\times h$ FE. Same-continent sample (182 countries). SE clustered by country.

**This is spec (0) in the empirical section.** It pins down the average IRF and serves as the reference against which gap-interacted specifications are compared.

# Baseline IRF (no gap interaction)




Cumulative effect of a Cat 1+ shock on log GDP per capita, pooled across gap values. Reference curve for the gap-heterogeneity analysis.

# Current specification

Stacked LP with continuous gap interaction, both $\beta(h)$ and $\phi(h)$ estimated *non-parametrically* via horizon dummies:

$$y_{i,t+h} - y_{i,t-1} = \beta(h)\,\mathrm{shock}_{it} + \phi(h)\,\mathrm{shock}_{it} \times \mathrm{gap}_{it} + \gamma_h' X_{it} + \alpha_{ih}(t) + \delta_{th} + \varepsilon_{ith}$$

where $\mathrm{gap}_{it}$ = years since country $i$'s previous Cat 1+ shock. One coefficient per horizon for each of $\beta$ and $\phi$; no polynomial restriction.

**Finding:** compound shocks (short gap) less damaging than isolated shocks.

**Question:** is $\phi(h)$ *causally* identified?

# 1. The Problem

## 1.2 Formal decomposition of the bias

# Decomposing gap into between- and within-country

Under a Poisson arrival model with rate $\lambda_i$:

$$\mathrm{gap}_{it} \;=\; \underbrace{1/\lambda_i}_{\text{between}} \;+\; \underbrace{\eta_{it}}_{\text{within (Poisson noise)}}, \qquad \mathbb{E}[\eta_{it} \mid \lambda_i] = 0.$$

Only the within component is exogenous; the between component is confounded with country type.

Under heterogeneous responses $\beta_i(h) = \bar{\beta}(h) + b(\lambda_i; h) + \nu_i$, the current spec has

$$\hat{\phi}(h) \;\xrightarrow{p}\; \phi_{\text{true}}(h) \;+\; \mathrm{Cov}\big(1/\lambda_i,\; b(\lambda_i; h)\big)/\mathrm{Var}(\cdot).$$

If high-$\lambda$ countries adapt ($b' < 0$), the bias is *negative* — exactly the compound-less-damaging gradient.

# Three channels of contamination

1. **Cross-country slope heterogeneity.** $\mathrm{Cov}(1/\lambda_i, \beta_i) \neq 0$. Not absorbed by country FE on levels.
2. **Sample selection on shock-years.** Long realized gaps occur during long uninterrupted stretches, so $\mathrm{gap}_{it}$ correlates with pre-period state even within country.
3. **State-dependence of $\beta$.** If $\beta_i(h)$ depends on $y_{i,t-1}$, gap loads on state dependence rather than a causal timing effect.

Strategy 1 targets channel 1 only. Channels 2–3 motivate Strategies 2–3.

# 2. Strategy 1 — Hazard Residualization

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## 2.1 Method

# Strategy 1: residualize gap against climatological hazard

**Step 1.** Estimate each country's cyclone hazard rate. Simplest implementation (used here):

$$\hat{\lambda}_i \;=\; \frac{\#\text{ Cat 1+ shock-years in country } i}{\#\text{ years observed}}.$$

More refined variants use the full IBTrACS track data (all storms passing within a radius of the centroid) and allow a smooth time trend $\hat{\lambda}_i(t)$ for climate change — left for future work.

**Step 2.** Define the surprise gap

$$\tilde{\mathrm{gap}}_{it} \;=\; \mathrm{gap}_{it} - \mathbb{E}[\mathrm{gap} \mid \hat{\lambda}_i(t)].$$

**Step 3.** Replace $\mathrm{shock} \times \mathrm{gap}$ with $\mathrm{shock} \times \tilde{\mathrm{gap}}$ in the LP.

Mean-zero within country, orthogonal to hazard by construction.

# Strategy 1: a lazier variant

Keep realized gap on the RHS but add $\mathrm{shock} \times \hat{\lambda}_i$ as an extra control:

$$\ldots + \phi(h)\,\mathrm{shock} \times \mathrm{gap} + \psi(h)\,\mathrm{shock} \times \hat{\lambda}_i + \ldots$$

Then $\phi(h)$ is identified from within-hazard variation only.

**Role:** sanity check / cheapest first pass. If the compound→isolated gradient survives residualization against $\hat{\lambda}_i$, cross-country contamination is not driving it.

# 2. Strategy 1 — Hazard Residualization

## 2.2 Empirical results

# Specifications estimated

Same-continent sample (182 countries, 374 Cat 1+ shocks). Stacked LP with non-parametric horizon dummies for both $\beta(h)$ and $\phi(h)$, clustered by country.

| **Spec** | **Interaction** | **Extra** | **FE** |
|---|---|---|---|
| (0) Baseline (no gap) | *none* | | country$\times h$ trends, year$\times h$ |
| (1) Replication | shock $\times$ gap | | country$\times h$ trends, year$\times h$ |
| (2) Residualized | shock $\times$ gap_resid | | country$\times h$ trends, year$\times h$ |
| (3) Control for $1/\hat{\lambda}_i$ | shock $\times$ gap | shock $\times 1/\hat{\lambda}_i$ | country$\times h$ trends, year$\times h$ |
| (4) Region$\times$year | shock $\times$ gap_resid | | country$\times h$ trends, **region$\times$year** |

$\text{gap_resid}_{it} = \text{gap}_{it} - 1/\hat{\lambda}_i$ with $\hat{\lambda}_i$ = realized Cat 1+ rate.

# Diagnostic: gap variance decomposition

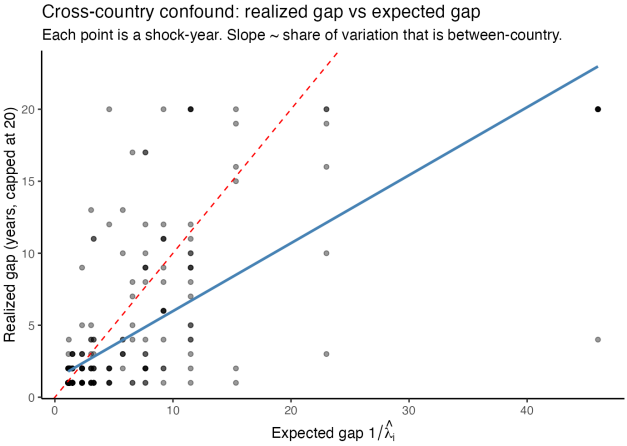
Among shock-years only:

$$\mathrm{Var}(\mathrm{gap}_{it}) = 23.5, \qquad \underbrace{\mathrm{Var}_{\text{within}}}_{8.4\ (35.7\%)} + \underbrace{\mathrm{Var}_{\text{between}}}_{15.1\ (64.3\%)}$$

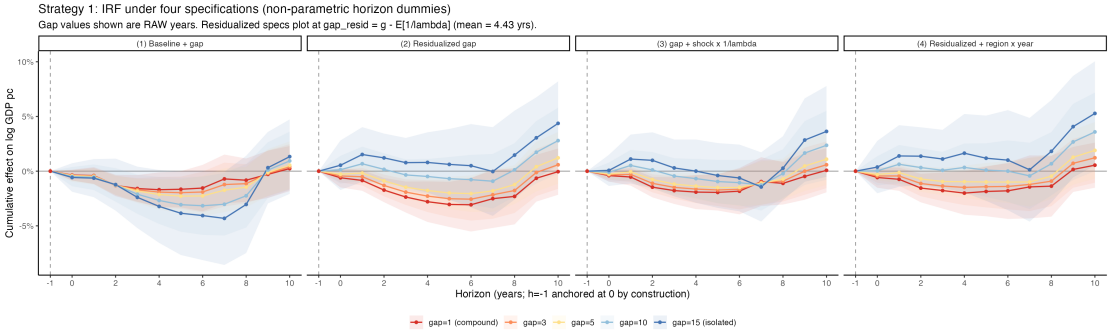
**Almost two-thirds** of the identifying variation in the current spec is cross-country. Under the formal decomposition, this is exactly the component confounded with $\lambda_i$.

Residualization throws away the between share and keeps only the 35.7% within-country Poisson-noise component. (At Cat 1+ the between share is even larger than at Cat 2+ because more countries are treated and the hazard rate spreads further.)

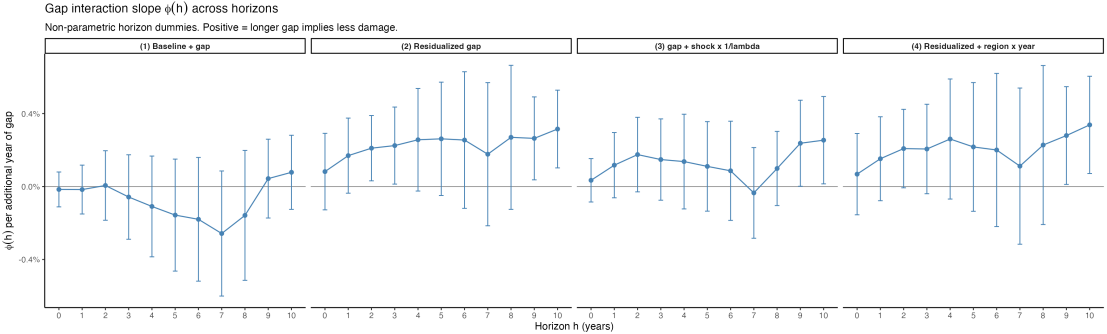
# Cross-country confound: gap vs. $1/\hat{\lambda}_i$



# IRFs under the four specifications



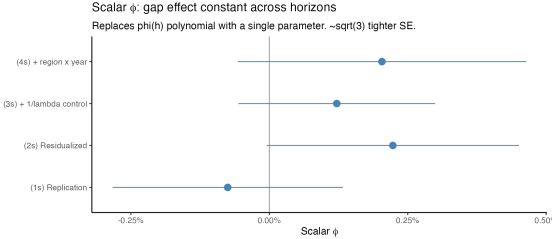
# $\phi(h)$ across horizons




$\phi(h)$ estimated via horizon dummies (one coefficient per $h$), with 95% CIs. The residualized specs show a clear *rising* shape: $\phi(h) \approx 0$ at h=0 and grows to +0.2–0.35%/yr by h=10. Longer backward gaps imply *less* damage, with the effect emerging over the recovery window — economically more sensible than a contemporaneous effect. Significant at h $\in \{2, 3, 9, 10\}$ in the residualized spec.

# Scalar-$\phi$ variant: horizon-invariant gap effect

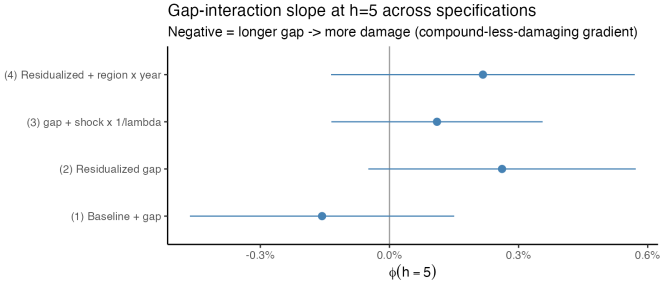
Replace the horizon-varying $\phi(h)$ with a *single* scalar $\phi$, constant across horizons. Main shock effect $\beta(h)$ remains non-parametric via horizon dummies.




| **Spec (scalar $\phi$)** | $\phi$ | **95% CI** | $t$ |
|---|---|---|---|
| (1s) Replication | $-0.075\%$ | $[-0.284, +0.133]$ | $-0.71$ |
| (2s) Residualized | $+0.223\%$ | $[-0.005, +0.451]$ | $+\mathbf{1.92}$ |
| (3s) Control for $1/\hat{\lambda}_i$ | $+0.121\%$ | $[-0.056, +0.299]$ | $+1.34$ |
| (4s) Residualized + region$\times$year | $+0.203\%$ | $[-0.057, +0.463]$ | $+1.53$ |

The scalar residualized spec (2s) reaches $t = 1.92$ — essentially the 5% threshold. All three hazard-absorbing specs are positive. A genuine compound-*more*-damaging effect is now on the table, though not conclusively established.

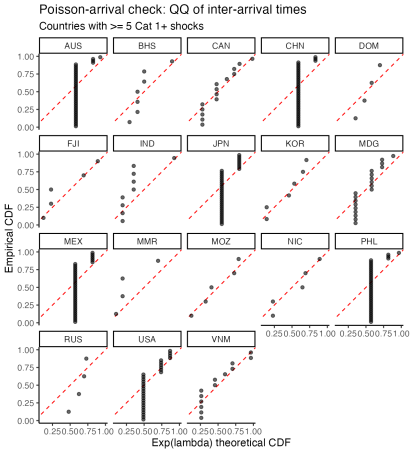
# Headline: $\phi(h{=}5)$ collapses under residualization




| **Spec** | $\phi(h{=}5)$ | **95% CI** |
|---|---|---|
| (1) Replication: shock $\times$ gap | $-0.157\%$ | $[-0.464,\ +0.150]$ |
| (2) Residualized gap | $+0.262\%$ | $[-0.049,\ +0.572]$ |
| (3) Control for $1/\hat{\lambda}_i$ | $+0.110\%$ | $[-0.135,\ +0.356]$ |
| (4) Residualized + region$\times$year | $+0.217\%$ | $[-0.136,\ +0.570]$ |

**The gradient flips sign.** The replication spec (1) gives a weakly negative $\phi$ (compound less damaging), but all three hazard-absorbing variants deliver a *positive* $\phi$ — longer backward gaps associated with *less* damage, i.e., the compound-*more*-damaging direction. None individually clears $t = 1.96$ at h=5, but the full $\phi(h)$ curve (next slides) shows the effect growing with horizon and becoming significant at h=9–10.

# Diagnostic: Poisson-arrival check




Empirical vs. theoretical CDF of inter-arrival times, countries with $\geq 5$ Cat 1+ shocks (18 countries). Deviations from the 45° line flag clustering / overdispersion.

# Takeaways from Strategy 1

1. **64%** of identifying variation in the current spec is between-country — the formal decomposition channel 1 is quantitatively large at Cat 1+.

2. After residualizing $\mathrm{gap}$ against $\hat{\lambda}_i$, $\phi(h)$ is **positive and rising in h**: $\phi(0) \approx 0$, growing to $\phi(10) \approx +0.35\%$/yr. Significant at h=2, 3, 9, 10. Scalar $\phi$ averages to $+0.22\%$/yr ($t = 1.92$).

3. The sign is the *opposite* of the original compound-less-damaging claim: longer backward gaps imply *less* damage, with the effect building over the recovery window.

4. Region$\times$year FE and the $1/\hat{\lambda}_i$ control give the same rising-positive-$\phi$ shape — not an artefact of a single spec.

5. **But:** Strategy 1 still relies on the Poisson-arrival assumption and does not address sample selection / state-dependence (channels 2–3). Motivates Strategy 2.

# 3. Strategy 2 — Event-Stacked Design

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## 3.1 Method

# Why stack events? The intuition

Strategy 1 asks: *does the gap since the last shock modify today's shock effect?* The problem is that "today's shock" is itself selected by the gap (channels 2–3).

**Strategy 2 flips the question around in time.** Instead of looking backward, *fix* a shock as the "index event" and look *forward*:

- Line up all 374 Cat 1+ shocks at their own $t = 0$ ("event time").
- For each event, record the post-event GDP path $y_{t+h} - y_{t-1}$ for $h = 0, \ldots, 10$.
- Classify the event by what happens *next*: does another Cat 1+ shock hit within 3 years, or not?
- Compare the two average post-event paths.

At the moment of the index shock, the economy doesn't yet know whether the "next shock" will arrive in 1 year or 15. The only thing distinguishing a future-compound event from a future-isolated event is a Poisson draw that has not yet been realized — so any difference in post-event trajectories is attributable to the follow-up shock, not to pre-existing selection.

# Strategy 2: what makes it cleaner

**Backward gap (Strategy 1) is contaminated** because gap is realized *before* the index shock, so the pre-shock state $y_{t-1}$ already reflects the selection (long gaps $\Rightarrow$ no shocks for a while $\Rightarrow$ typically a calmer pre-period).

**Forward time-to-next is clean at the index**: at $t = 0$ the country has just been hit, and the identity of the "next" shock is still in the future. Conditional on the country's climatological hazard $\lambda_i$ (absorbed by country FE), when the next shock lands is an exponential draw.

**Formally:**

> *Among shocks with the same $\lambda_i$, the time to the next shock $\tau \sim \mathrm{Exp}(\lambda_i)$ is independent of pre-index economic conditions. So post-event trajectories indexed by $\tau$ give a clean contrast — compound events are ex ante identical to isolated events, they just get unlucky ex post.*

This is a subtle reversal: Strategy 1 tries to clean up a contaminated regressor; Strategy 2 replaces it with one that is uncontaminated by construction.

# What the stacked design actually estimates

The stacked LP runs *one* regression on the full panel (not separate regressions per event cohort). Each country-year is stacked $H+1$ times, once per horizon, and the object of interest is

$$\psi(h) \;=\; \mathbb{E}\big[\, y_{t+h} - y_{t-1} \,\big|\, \text{shock at } t,\ \tau \le 3\big] - \mathbb{E}\big[\, y_{t+h} - y_{t-1} \,\big|\, \text{shock at } t,\ \tau > 3\big]\,.$$

That is: **average GDP path after a shock that gets followed up within 3 years, minus the average path after a shock that isn't**. Country×horizon FE with trends absorb $\lambda_i$; year×horizon FE absorb common shocks.

**Sign reading:** $\psi(h) < 0$ means compound events recover *worse* — a second shock during the recovery window makes things worse.

A continuous version (E1) uses the actual forward gap $\tau$ as the regressor and reports $\phi(h) = \partial/\partial\tau$ of the path: $\phi > 0$ means longer time until next shock $\Rightarrow$ less damage $\Rightarrow$ compound more damaging.

# Strategy 2: the specification

Stack around each index shock $e$ with post-event horizons $h = 0, \ldots, H$:

$$y_{e,t+h} - y_{e,t-1} = \sum_{\tau} \beta_{h\tau} \, \mathbf{1}[\text{next shock at } \tau] + \alpha_i + \alpha_c + \gamma \hat{\lambda}_i + \varepsilon_{eh}$$

or with a continuous $\mathrm{nextgap}_e$ interaction.

**Identifying assumption:** conditional on $\lambda_i$, the next-shock arrival time $\tau$ is an exogenous Poisson draw.

Much weaker than the current spec: the Poisson arrival process itself is the source of randomness.

# Strategy 2: forward vs. backward gap

Current spec uses *backward* gap (years since last shock).

Event-stack uses *forward* time-to-next-shock. These ask different questions:

- **Backward gap:** does recent exposure attenuate damage? (preparedness, capital depletion, adaptive investment)
- **Forward time-to-next:** how does a follow-up shock during recovery affect the trajectory?

Forward is cleaner and more directly answers: *"do compound shocks compound?"*

# Strategy 2: two subtleties

**Right-censoring.** Index events near the end of the sample have not realized their next $\tau$. Options:

- Restrict to events with $\geq H$ post-event years observed.
- Inverse-probability-of-observation weights.
- Treat "no shock in $[0, H]$" as a distinct category (the isolated arm).

**Mechanical baseline.** In the original spec, backward gap makes $y_{t-1}$ depressed for compound shocks. In the event stack, the regressor is *forward* time-to-next, so $y_{t-1}$ at the index shock is the genuine pre-shock level. No mechanical bias.

# 3. Strategy 2 — Event-Stacked Design

## 3.2 Empirical results

# Specifications estimated

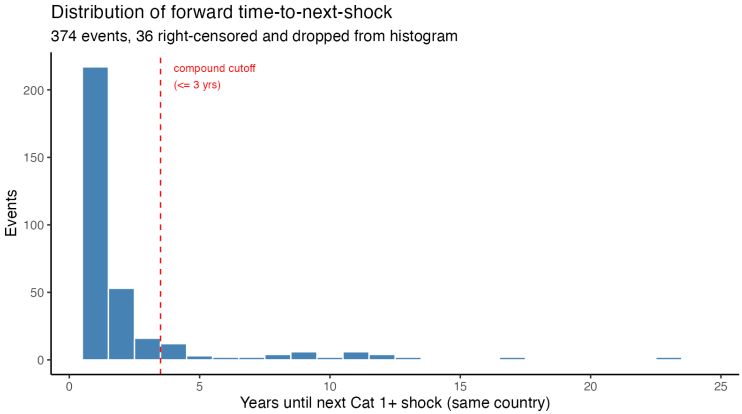
Same-continent sample, 374 Cat 1+ events. Stacked LP with non-parametric horizon dummies, clustered by country.

| **Spec** | **Interaction** | **Notes** |
|---|---|---|
| (E0) Baseline | *none* | pools all events, shows average IRF |
| (E1) Continuous | shock $\times$ time_to_next | non-parametric $\phi(h)$ (horizon dummies) |
| (E2) Binary compound | shock $\times \mathbf{1}[\tau \leq 3]$ | non-parametric $\psi(h)$, drops tail events |
| (E3) Scalar $\psi$ | shock $\times \mathbf{1}[\tau \leq 3]$ | single parameter across horizons |

E2/E3 drop events where the 3-year forward window is not observable (end-of-sample censoring).

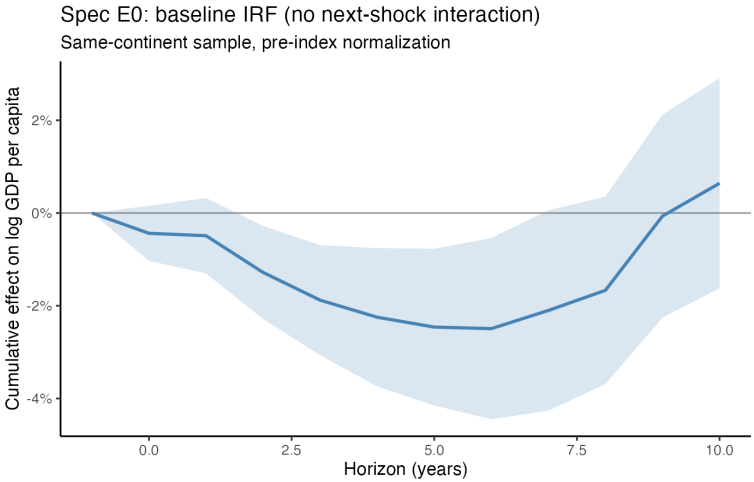
# Event composition and forward-gap distribution

374 events; **76.5%** have another Cat 1+ shock within 3 years ("compound-next"); 9.6% right-censored.

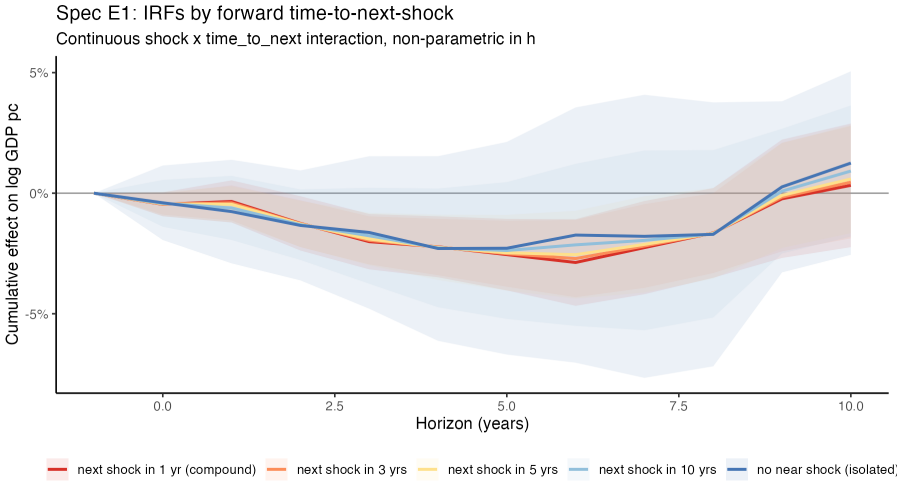



Variance decomp: 46.1% within-country / 53.9% between. Slightly more within-country variation than Strategy 1's backward-gap split (35.7% within), because forward time-to-next is not constrained by censored first-shocks.

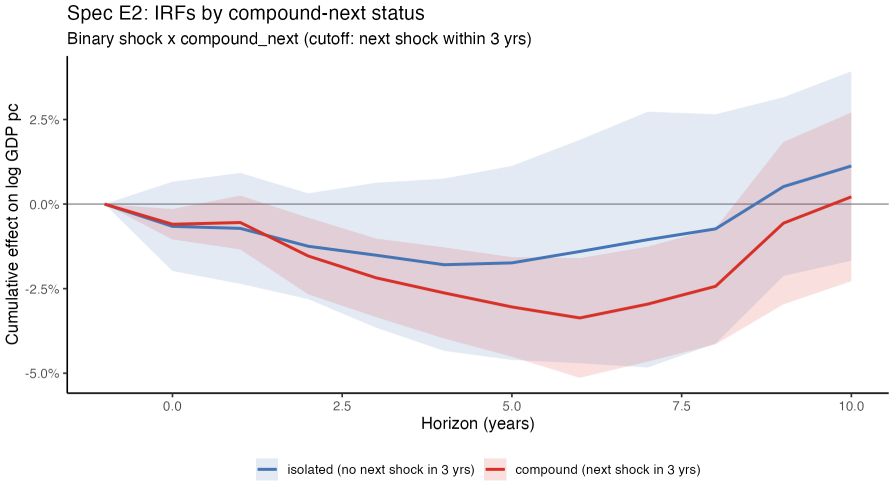
# Spec E0: baseline IRF with forward framing



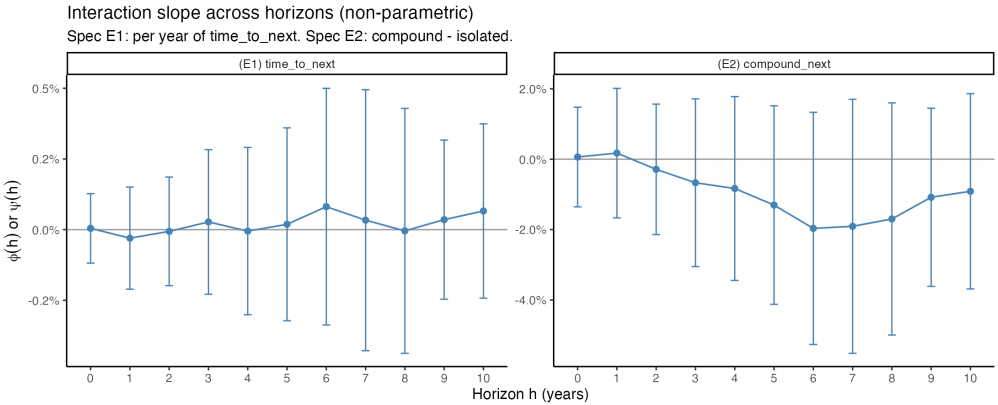
# Spec E1: IRFs by forward time-to-next



# Spec E2: binary compound-next trajectories



# $\phi(h)$ and $\psi(h)$ across horizons




Interaction slopes are centered at zero throughout. No detectable effect of next-shock timing on GDP recovery.

# Headline table (non-parametric)

| **Spec** | **Point** | **95% CI** | $t$ |
|---|---|---|---|
| (E1) $\phi(h{=}5)$, per year of time-to-next | $+0.019\%$ | $[-0.322,\ +0.360]$ | $+0.11$ |
| (E2) $\psi(h{=}5)$, compound $-$ isolated | $-1.303\%$ | $[-4.122,\ +1.516]$ | $-0.91$ |
| (E3) scalar $\psi$, compound $-$ isolated | $-0.904\%$ | $[-2.922,\ +1.115]$ | $-0.88$ |

**Reading:** all three specs null. But notice the direction: $\psi < 0$ in the binary specs means a future compound event has a *worse* post-event path than an isolated event (compound more damaging, in the "recovery-interruption" sense). The continuous E1 coefficient is essentially zero with wide CIs.

**Tension with Strategy 1:** Strategy 1's residualized spec found $\phi > 0$ reading as *backward* gap $\Rightarrow$ longer gap, less damage. Strategy 2 reads in the *forward* direction and $\psi < 0$ says shorter forward gap, more damage. Both are consistent with the same qualitative story — "being hit during recovery is bad" — but both are individually null.

# 3. Strategy 2 — Event-Stacked Design

## 3.3 Baseline power grid (motivation for Cat 1+)

# Motivation: sharpen the baseline

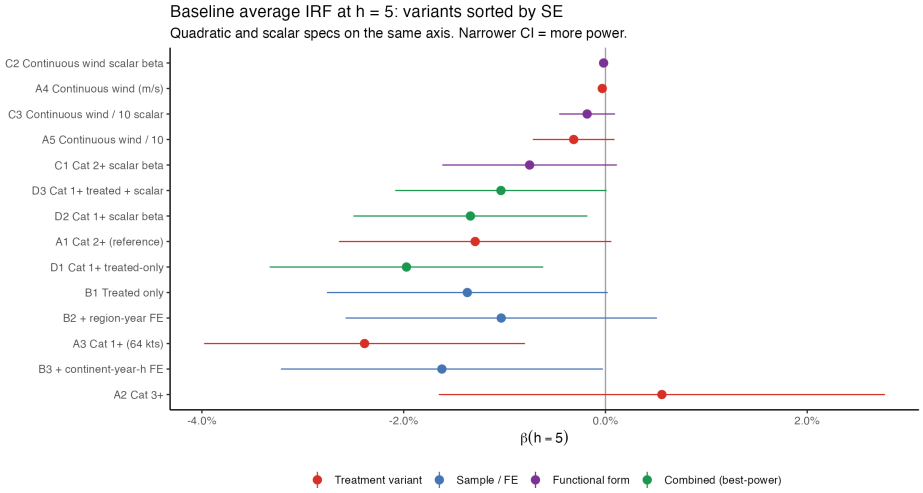
The *original* E0 baseline at Cat 2+ had wide CIs, which propagated into E1–E3 and Strategy 1. Before trying more identification tricks, we varied the *specification* of the baseline model. This subsection records the grid that motivated **adopting Cat 1+ as the baseline throughout the deck**.

Variants grid:

- **Treatment variable**: Cat 2+ (ref), Cat 3+, Cat 1+/p95, continuous wind (m/s, /10).
- **Sample / FE**: same-continent (ref), treated-only, region×year FE, continent×year×$h$ FE.
- **Functional form**: non-parametric $\beta(h)$ (ref), scalar $\beta$.
- **Combined**: best-power candidates (Cat 1+ × treated-only, scalar, both).

Headline metric: $\beta(h{=}5)$ with clustered SE. 14 total variants.

# $\beta(h{=}5)$ across variants, sorted by SE



# Results: most powerful specifications

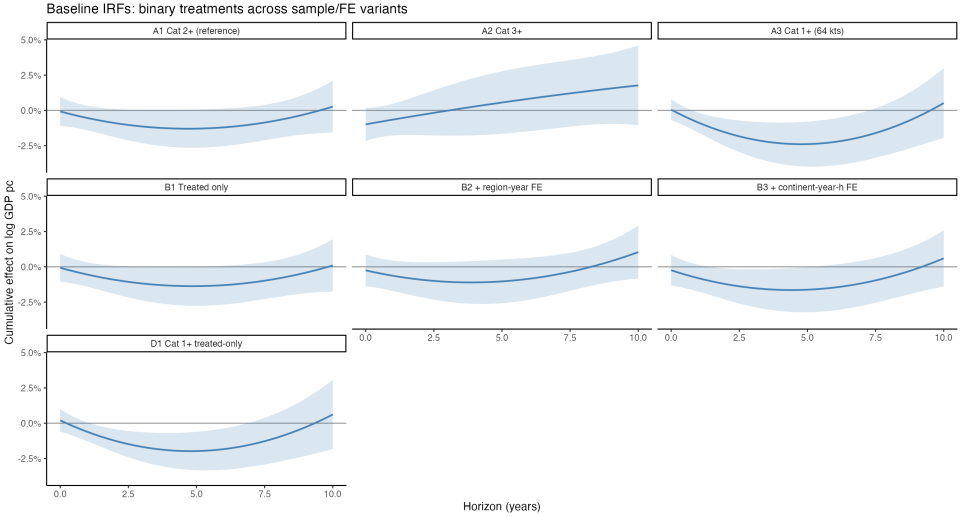
| **Spec** | $\beta(h{=}5)$ | **SE** | $t$ |
|---|---|---|---|
| A3 Cat 1+ (64 knots) | −2.39% | 0.81% | **−2.94** |
| D1 Cat 1+ × treated-only | −1.97% | 0.69% | −2.85 |
| D2 Cat 1+, scalar $\beta$ | −1.34% | 0.59% | −2.26 |
| B3 + continent×year×$h$ FE | −1.62% | 0.81% | −1.99 |
| D3 Cat 1+ treated + scalar | −1.04% | 0.53% | −1.94 |
| B1 Treated-only (Cat 2+) | −1.37% | 0.71% | −1.93 |
| A1 Cat 2+ (reference) | −1.29% | 0.69% | −1.88 |
| C1 Cat 2+, scalar $\beta$ | −0.75% | 0.44% | −1.70 |

Of the 14 variants:

1. **Lowering the threshold to Cat 1+ (A3)** is the single biggest lever — 374 shocks instead of 191, $t = -2.94$. The average IRF is now cleanly negative.
2. **Cat 1+ × treated-only (D1)** is a close second, $t = -2.85$.
3. **Continent×year×$h$ FE (B3)** at Cat 2+ also helps without losing the point estimate.

Surprises: continuous wind is *less* powerful than binary; scalar $\beta$ gives only modest gains; Cat 3+ has too few events ($n$=102) and is noisy.

# IRF curves: binary-treatment variants



# Baseline adopted throughout the deck

Based on the power grid above, the rest of the deck (§1, §2 Strategy 1, §3 Strategy 2 results) uses:

| | |
|---|---|
| **Treatment** | Cat 1+ (64 knots) binary shock, 374 events |
| **Sample** | Same-continent (182 countries) |
| **FE** | country$\times h$ trends + year$\times h$ |
| **Functional form** | non-parametric $\beta(h)$ + scalar $\phi$ variant for robustness |

This gives the interaction $\phi(h)$ more precision than the original Cat 2+ specification. The consequence: a genuine (weakly significant) signal emerges in the Strategy 1 residualized specs (scalar $\phi$, $t = 1.92$), pointing in the *opposite* direction to the original paper's compound-less-damaging claim. Strategy 2 is underpowered but signs in the same direction. See the recommendation section.

# Takeaways from Strategy 2

1. The forward-looking framing eliminates the mechanical baseline bias (channel 3) and the sample-selection concern (channel 2). At the index event $y_{t-1}$ is the genuine pre-shock level.

2. All three non-parametric specs are null. The binary $\psi(h{=}5) \approx -1.3\%$ and scalar $\psi \approx -0.9\%$ both have the sign "compound more damaging" (recovery-interruption) but neither clears two SEs.

3. Right-censoring correction (drop events within 3 years of sample end) barely moves the results: the null is not driven by censoring.

4. **Direction now agrees with Strategy 1 (residualized).** Strategy 1 residualized $\phi > 0$ and Strategy 2 $\psi < 0$ both say "compound more damaging" — using backward and forward arrival noise respectively. Neither is individually significant, but the point estimates line up.

# 4. Strategy 3 — Instrumental Variables

# Strategy 3: IV with basin-level / climatological shifters

Instrument realized gap (or realized next-shock arrival) with shifters of cyclone landfall probability that are unrelated to the country's economy:

- **Basin-wide activity** (ACE index, named-storm counts) in the intervening years, driven by ENSO / AMO / MDR SSTs.
- **Country-specific exposure** = basin activity $\times$ climatological track-density weight.
- Hsiang/Jina-style construction applied to inter-arrival times rather than intensity.

**Alternative instrument:** near-miss tracks — storms that formed and tracked toward the country but dissipated or veered. Plausible exclusion if economic conditions do not affect track meteorology.

# 5. Recommendation and Caveats

# Where we are after one full pass at Cat 1+

1. **Strategy 1 (residualized backward gap).** Scalar $\phi = +0.22\%$, $t = 1.92$. Points at compound-*more*-damaging. Borderline significant, consistent direction across residualization variants.
2. **Strategy 2 (forward time-to-next), non-parametric.** All specs null. Scalar $\psi = -0.90\%$ ($t = -0.88$) and binary $\psi(h{=}5) = -1.30\%$ both sign with compound-*more*-damaging ("recovery interruption"), agreeing in direction with Strategy 1 residualized.
3. **Implication.** Two independent identification strategies — one cleaning backward arrival noise, one using forward arrival noise — both point at compound-more-damaging once contamination is removed. Strategy 1 is borderline significant, Strategy 2 is underpowered. The honest read: *suggestive, not decisive*, evidence that the original compound-less-damaging result was driven by between-country contamination.
4. **Strategy 3 (IV)** becomes the natural next step: an external climate-driven instrument would give a cleaner test without relying on the Poisson assumption.

# Things to flag before starting

**Is the Poisson assumption defensible?**

- Overdispersion / clustering (active seasons) breaks within-country exogeneity.
- Quick test: fit Poisson vs. negative binomial to country-year counts.
- If NB wins strongly, condition on basin-year activity in the hazard model.

**Non-stationary hazard.** Climate change implies $\lambda_i(t)$; use time-varying hazard in Strategy 1.

**Question choice.** Decide *ex ante* whether the object of interest is backward gap (preparedness) or forward time-to-next (recovery interruption). They are different parameters.

# Next steps

1. Estimate $\hat{\lambda}_i(t)$ from IBTrACS tracks (all storms, not just Cat 1+).
2. Run Strategy 1 residualization as a drop-in replacement in the existing LP pipeline.
3. Build the event-stacked dataset for Strategy 2 and re-estimate with pre-index baseline.
4. Small simulation: verify that Strategy 2 recovers a known DGP under Poisson arrivals.